MS1-Based Workflow for the Identification of Plastic Oligomers and Non-Intentionally Added Substances (NIAS)

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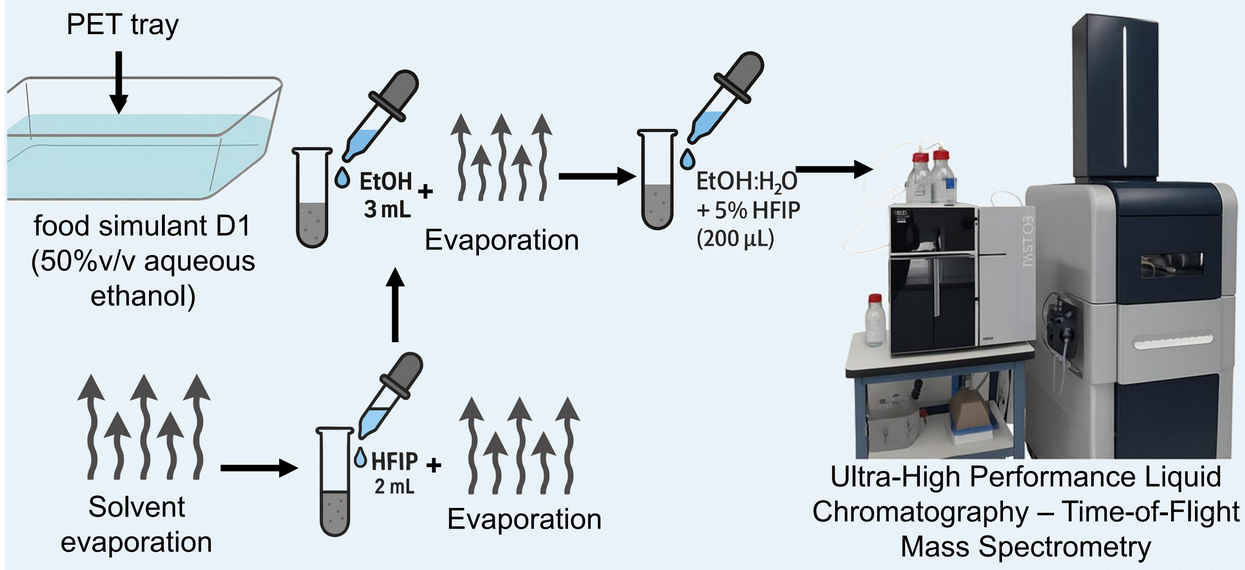
Introduction

Polyethylene terephthalate (PET) is widely used in food packaging, yet concerns persist over the migration of non-intentionally added substances (NIAS)—especially from recycled PET (rPET) [1]. In this study, migration behavior of PET oligomers was evaluated using food simulant D1 (50% v/v ethanol) in virgin and rPET trays, with high-resolution analysis conducted via UHPLC-TOF-MS.

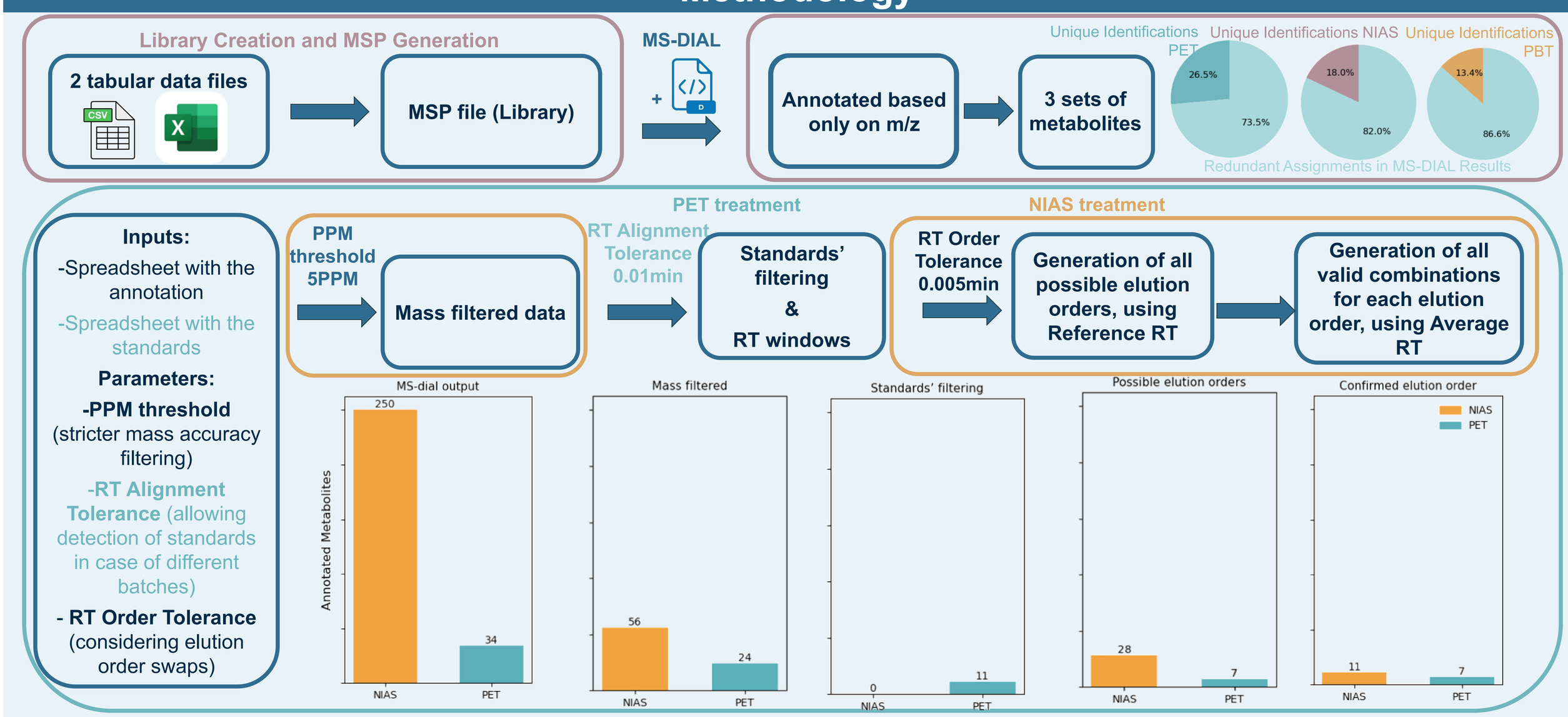
To aid compound identification, two literature-based spectral libraries[2], [3] —focusing on PET-related migrants and known NIAS—were applied. The approach is grounded in the principle that in reversed-phase (RP) chromatography, compounds tend to maintain their relative elution order, even across varying conditions (e.g., gradient, column, or mobile phase).

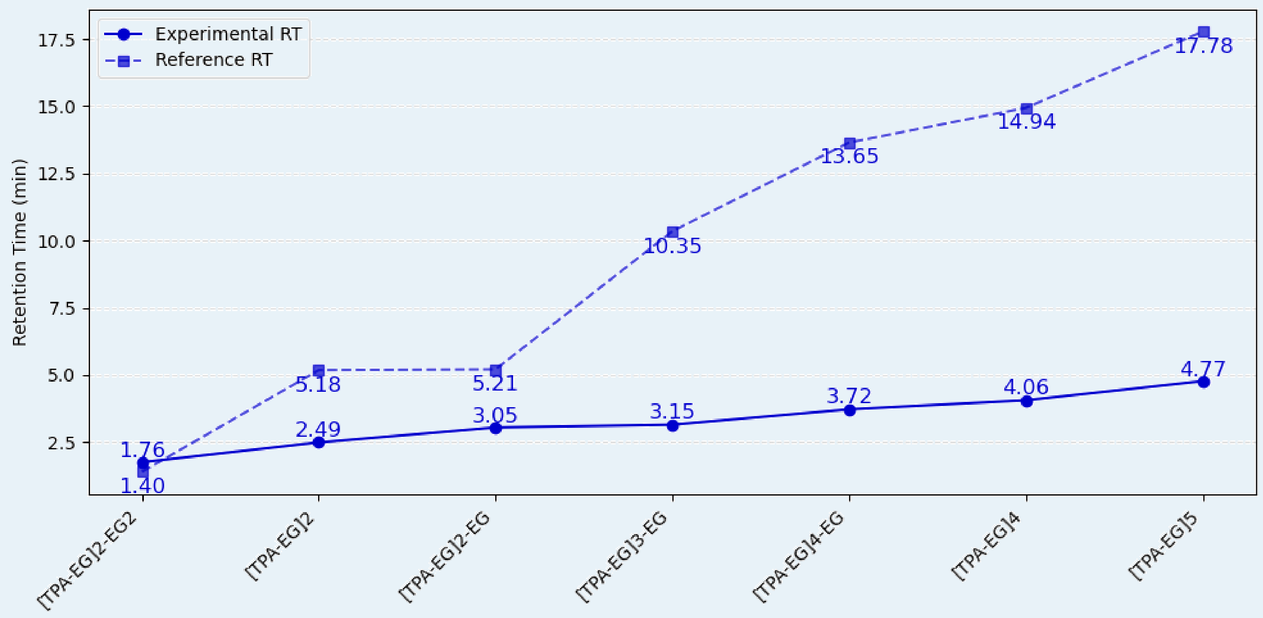
Leveraging this, two MS1-level annotation algorithms were developed: one integrating retention time windows from reference standards, and another relying solely on elution sequence. This enabled the creation of a curated RP-detectable library of PET-related compounds and a reproducible, adaptable identification workflow applicable across other compound classes analyzed via RP chromatography.

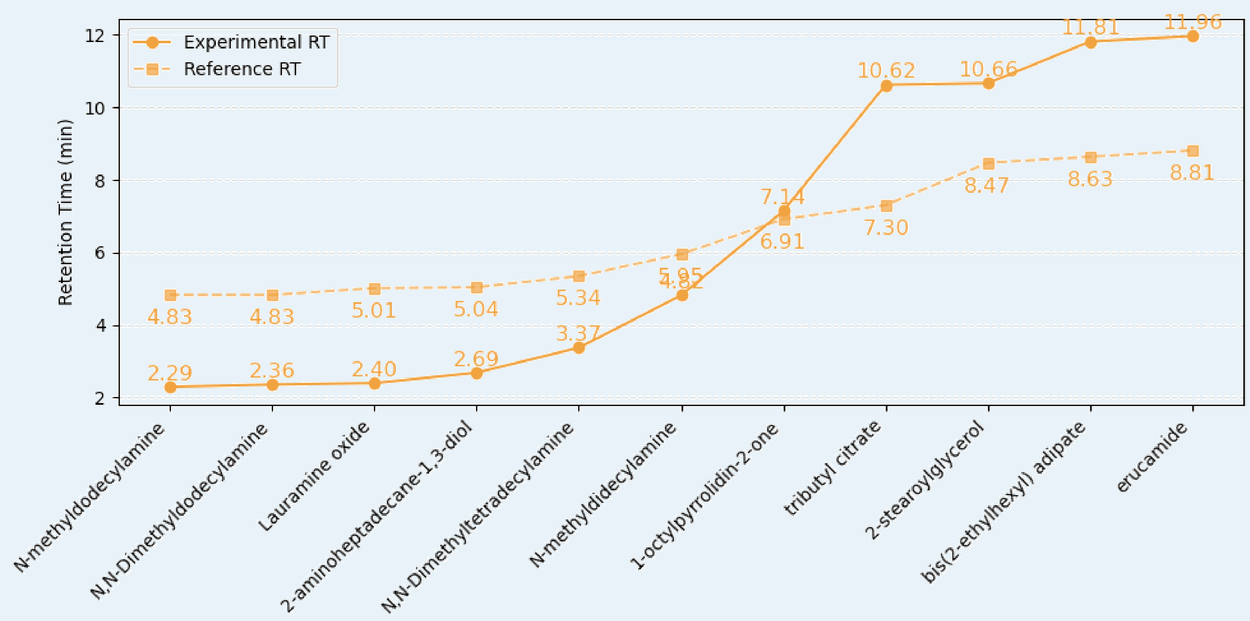
Experimental



Methodology







Access Workflow Resources

Scan to access all supporting materials, including algorithms, MSP library that was used and documentation used in this study.

Conclusions

- An MS1-based workflow was established for the identification of plastic oligomers and NIAS. High annotation redundancy revealed the limitations of m/z-based identification alone, highlighting the need for integrating retention time (RT) and MS/MS data to achieve accurate and specific compound annotation.
- Ready-to-use algorithms were developed to allow the broader application of this annotation strategy with any MSP library derived from published literature.
- A unified MSP library was also created, incorporating PBT compounds annotated through in-house standard measurements.

References

[1] G. Colombo, M. Corredig, I. Uysal Ünalan, E. Tsochatzis, Untargeted screening of NIAS and cyclic oligomers migrating from virgin and recycled polyethylene terephthalate (PET) food trays, Food Packaging and Shelf Life 41 (2024) 101227. https://doi.org/10.1016/j.fpsl.2023.101227.E.

[2] D. Tsochatzis, J. Alberto Lopes, O. Kappenstein, T. Tietz, and E. J. Hoekstra, "Quantification of PET cyclic and linear oligomers in teabags by a validated LC-MS method – In silico toxicity assessment and consumer's exposure," *Food Chem*, vol. 317, p. 126427, Jul. 2020, doi: 10.1016/j.foodchem.2020.126427.

[3] Q. Z. Su, P. Vera, and C. Nerín, "Combination of Structure Databases, In Silico Fragmentation, and MS/MS Libraries for Untargeted Screening of Non-Volatile Migrants from Recycled High-Density Polyethylene Milk Bottles," *Anal Chem*, vol. 95, no. 23, pp. 8780–8788, Jun. 2023, doi: 10.1021/acs.analchem.2c05389.